EXP 9:: SUBNETTING

AIM:

Implementation of subnetting in CISCO PACKET TRACER simulator.

INTRODUCTION:

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. Classless subnetting allows us to use subnet masks other than the default class-based masks, making IP addresses more efficient and allowing multiple networks to be created.

Here the n/w 192.168.1.0/24 is divided into smaller /27 subnets; provides 30 usable host addresses per subnet.

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Add required routers, switches and PCs.
3. Connect the devices using appropriate tables.
4. Configure the routers with the required IP addresses and subnet masks.
5. Configure the router interfaces using the CLI:
   - enable
   - configure terminal
   - select the required Fast Ethernet / GigabitEthernet
   - assign the IP address and subnet masks.
   - Use no shutdown to activate the interface.

6. Configure the switch ports in access mode where required.

7. Assign static IP addresses, subnet mask and default gateways to the PC's.

8. Use the /27 subnet mask : 255.255.255.

9. Configure the four subnets:

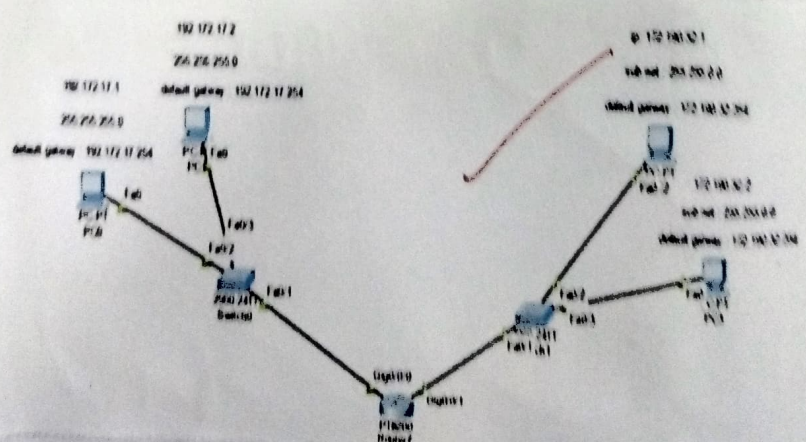
   - 192.168.1.0/27
   - 192.168.2.0/27
   - 192.168.3.0/27
   - 192.168.4.0/27

10. Verify the router interfaces and PC's belong to their respective routers.

11. Open command prompt in the PC's.

12. Use the ping command to test connectivity between PC and respective router.

Subnetting Network Topology:



Ping used to ~~very~~ verify communication between PC's in different subnets:

RESULT:

Thus, subnetting was successfully implemented using CISCO PACKET TRACER.